

Julia v1.5 SMC²-MPC

Filter HMC unified with the framework’s ChEES path

3-horizon test sweep, $T \in \{14, 28, 42\}$ d, RTX 5090, seed 42

2026-05-09

Abstract

This report records a 3-horizon test sweep of the FSA v1.5 closed-loop SMC²-MPC bench after two filter-HMC fixes: (i) the bench’s filter HMC was replaced with the framework’s generic `parallel_hmc_one_move_generic!` + `chees_pick_L_generic` (the same code path the controller uses), passing `beta = next_λ` so the kernel correctly targets the tempered intermediate; (ii) the HMC step size and finite-difference step size were updated to the sweet-spot values identified in the standalone HMC accept-rate sweep ($\varepsilon = 0.005$, $h_{fd} = 0.01$). All filter-side rejuvenation tricks (Liu–West $a = 0.97$, Silverman+KDE+LW `smooth_resample_bw=1.0`, Gaussian bridge between rolling windows) and the controller’s reduced “fast” configuration (`--ctrl-n-smc 1024 --ctrl-num-mcmc 8`) match the prior no-filter-HMC baseline so the comparison is apples-to-apples on the filter side. Per-tempering-level filter HMC diagnostics are now recorded by parsing the bench log; controller-HMC diagnostics continue to be recorded via the previous CSV instrumentation. All numerical entries in this report are pulled directly from CSVs on disk; the parameter-recovery reading in §10 is from the trace plots in §6 and §7.

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1 Purpose

The prior no-filter-HMC sweep (`compare_v15_julia_vs_python/example_run/julia_horizon_sweep_2026-05-09_no_filter_hmc/HMC_NOT_NEEDED_FINDINGS.md`) established a baseline: run the bench with filter HMC *disabled* (`--num-mcmc 0`) and rely on the rejuvenation tricks alone (Liu–West, Silverman+KDE, Gaussian bridge). Inspection of that sweep’s parameter traces showed several parameters locked at biased values across all six horizons.

This sweep tests whether re-enabling the filter HMC – after fixing two bugs in it – improves parameter recovery while keeping everything else identical.

Bug A (HMC step size ε and FD step h_{fd}). The standalone HMC accept-rate sweep at `compare_v15_julia_vs_python/example_run/julia_horizon_sweep_2026-05-09/diag_hmc_accept_sweep.csv` reported 0% acceptance at the bench’s defaults ($\varepsilon = 0.05$, $L = 4$, $h_{fd} = 10^{-3}$) and 50–74% at the sweet spot ($\varepsilon \in [10^{-3}, 10^{-2}]$, $h_{fd} = 10^{-2}$).

Bug B (λ -tempering missing on the MCMC kernel). The bench’s local `tempered_grads` closure inside `models/fsa_high_res/gpu_pf.jl::parallel_hmc_one_move!` returned the full posterior log-density and gradient regardless of the tempering level λ . Standard tempered SMC requires the MCMC kernel at level λ to leave $\pi_\lambda \propto p(\theta)p(y|\theta)^\lambda$ invariant, which means scaling the data log-likelihood by λ . Bug B was masked by Bug A: with 0% acceptance, no MCMC move was applied and the cloud stayed at π_λ by accident.

The fix ports the filter to the framework’s generic ChEES HMC. The controller side already had:

- `parallel_hmc_one_move_generic!` in `julia/SMC2FC_functional/src/Control/GPUControlSMC.jl`, which takes a `beta::Float64` keyword and computes `vals_prior + beta * vals_data`, fixing Bug B by construction.
- `chees_pick_L_generic`, which adapts the leapfrog count L per tempering level by maximising expected squared jump distance per $\varepsilon \cdot L$.

The bench’s filter call site was rewritten to call those framework functions, with `beta = next_λ`. The model-specific `parallel_hmc_one_move!` in `gpu_pf.jl` is now unused.

2 What was run

Per horizon $T \in \{14, 28, 42\}$, identical except `--T-days`:

```
julia --project=. tools/bench_smc_full_mpc_fsa_gpu.jl \  
  --T-days <T> --seed 42 \  
  --num-mcmc 3 \  
  --hmc-step-size 0.005 \  
  --hmc-leapfrog 4 \  
  --h-fd 0.01 \  
  --filter-chees-min 4 --filter-chees-max 64 \  
  --liu-west-a 0.97 \  
  --smooth-resample-bw 1.0 \  
  --gaussian-bridge true \  
  --N-smc 512 --K-per-chain 1000 \  
  --ctrl-n-smc 1024 --ctrl-num-mcmc 8 \  
  --collect-ctrl-diagnostics true \  
  --output-dir <SWEEP_ROOT>/T<T>d_seed42
```

Driven by the new sibling launcher `version_1.5_Julia/tools/launchers/run_julia_horizon_sweep_filter_hmc.unified.sh`.

2.1 Decoding each filter-side flag

All flag values listed below are the new bench defaults (2026-05-09): the bench’s `_parse_args` defaults dict in `version_1.5_Julia/tools/bench_smc_full_mpc_fsa_gpu.jl` has been updated so that a bare invocation (no flag overrides beyond `--T-days` and `--seed`) reproduces this run’s config. The four defaults that changed in this update: `--hmc-step-size` $0.05 \rightarrow 0.005$, `--smooth-resample-bw` $0.0 \rightarrow 1.0$, `--ctrl-n-smc` $2048 \rightarrow 1024$, `--ctrl-num-mcmc` $16 \rightarrow 8$. (The other flags `--num-mcmc` 3, `--hmc-leapfrog` 4, `--h-fd` 0.01, `--filter-chees-min` 4, `--filter-chees-max` 64, `--liu-west-a` 0.97, `--gaussian-bridge` true were already at these values after earlier 2026-05-09 edits.) To recover the original saturated-controller default pass `--ctrl-n-smc 2048 --ctrl-num-mcmc 16`.

- `--num-mcmc 3` re-enables the per-tempering-level filter HMC. Three moves per level (matches the saturated default).
- `--hmc-step-size 0.005` replaces the bench default of 0.05; from the standalone HMC sweep this is in the 50–75 % accept band.
- `--hmc-leapfrog 4` is the picker’s initial-fallback L ; the actual L per tempering level is selected from the candidate list [4, 8, 16, 32, 64] by ChEES.
- `--h-fd 0.01` replaces the previously-hardcoded $h_{fd} = 10^{-3}$ inside `filter_cfg`; this is the sweet-spot from the HMC accept-rate sweep.
- `--filter-chees-min 4 --filter-chees-max 64` build the ChEES candidate list as powers of 2: [4, 8, 16, 32, 64]. Shorter trajectories than the controller’s [16, 32, ..., 512] because each filter leapfrog step calls into the inner PF.
- Remaining filter rejuvenation flags (`--liu-west-a 0.97`, `--smooth-resample-bw 1.0`, `--gaussian-bridge true`) match the no-filter-HMC baseline so the only on-the-filter change is HMC.
- `--ctrl-n-smc 1024 --ctrl-num-mcmc 8` is the *reduced* (“fast-controller”) particle / MCMC budget, same as the no-filter-HMC baseline.

3 Wall and closed-loop metrics

Source: `horizons_results.csv` in this sweep’s root, written by `compare_v15_julia_vs_python/src/extract_hor`

T (d)	wall (s)	$\bar{A}_{\text{MPC}}$	$A_{\text{MPC}}^{\text{end}}$	$\bar{A}_{\text{base}}$	$A_{\text{base}}^{\text{end}}$	F violations
14	520.2	0.076	0.084	0.114	0.182	0
28	1148.2	0.090	0.155	0.181	0.342	0
42	1886.3	0.163	0.554	0.285	0.646	0

Table 1: Per-horizon wall, mean and final autonomic amplitude A under MPC and the constant- $\Phi = 1$ baseline rollout, fatigue-barrier ($F_{\text{max}} = 0.40$) violation count. Zero F violations across all three horizons.

3.1 Walls vs the no-filter-HMC sibling sweep

The no-filter-HMC sweep (`julia_horizon_sweep_2026-05-09_no_filter_hmc/`) ran the same horizons with `--num-mcmc 0` (filter HMC off) and all other flags identical. Wall ratios:

3.2 Closed-loop A trajectory matches the no-filter-HMC run

The closed-loop A -trajectory metrics are within numerical noise between the two runs (Table 3). The controller’s behaviour is dominated by its own configuration; the filter-side HMC change does not visibly disturb the closed-loop trajectory at this scale.

T (d)	this run (s)	no-filter-HMC (s)	ratio
14	520.2	133.2	$3.91\times$
28	1148.2	342.0	$3.36\times$
42	1886.3	715.5	$2.64\times$

Table 2: Wall is $2.6\text{--}3.9\times$ longer with the unified ChEES filter HMC than with the filter HMC disabled. The overhead is the new per-tempering-level cost: ChEES picker (one HMC move at each of 5 candidate L ’s on a small subset) plus `num_mcmc=3` production HMC moves at the chosen L .

T (d)	$\bar{A}_{\text{MPC}}$ this run	$\bar{A}_{\text{MPC}}$ no-HMC	$A_{\text{MPC}}^{\text{end}}$ this run	$A_{\text{MPC}}^{\text{end}}$ no-HMC
14	0.076	0.077	0.084	0.085
28	0.090	0.095	0.155	0.161
42	0.163	0.167	0.554	0.521

Table 3: Closed-loop A trajectory comparison.

4 Filter HMC measurements (NEW)

Per-tempering-level filter HMC accept rates and ChEES-picked L are parsed from each horizon’s `bench.log` via the new `compare_v15_julia_vs_python/src/diag_filter_hmc_accept.jl` script.

Per-horizon CSVs at `docs/filter_hmc_accept-T<N>d.csv`; cross-horizon summary at `docs/filter_hmc_accept_s`

T (d)	n levels	mean accept (%)	median (%)	min (%)	max (%)	mean ChEES L	mean levels / stride
14	130	71.35	73.00	24	94	58.22	5.00
28	270	65.47	68.00	13	94	54.70	5.00
42	410	61.90	64.00	10	93	51.84	5.00

Table 4: Filter HMC accept rates and ChEES-picked leapfrog count L , per horizon. Compare with the prior baseline’s measured **0 %** acceptance across 2880 logged tempering levels: the fix moves the filter HMC from rejecting every proposal to a textbook MCMC operating point (50–80 % accept).

ChEES selects L from the candidate list $\{4, 8, 16, 32, 64\}$; the reported mean $\sim 52\text{--}58$ indicates the picker is choosing near the top end of the list at most levels — consistent with $\varepsilon = 0.005$ allowing relatively long trajectories before the MH step rejects.

5 Controller HMC measurements

The controller HMC (already correctly β -tempered before this fix) was re-recorded for completeness via the existing `controller_diagnostics.csv` pipeline.

6 Posterior parameter traces (per horizon)

Figures 1–3: per-horizon posterior parameter trace plots (the bench’s internal `_plot_param_traces_v15` output). 10-panel layout per figure; per panel: posterior median (blue), 5–95 % credible band (light blue fill), truth value (red dashed), x -axis in days.

T	replans	total levels	mean accept (%)	median (%)	mean L	mean $\beta_{\max}$	mean ESJD/ εL	mean $\Delta \log D$	mean w
14	13	69	96.87	97.61	16	226.39	14.12	0.010	
28	27	147	95.43	96.68	16	27.02	13.73	0.085	
42	41	238	96.31	96.60	16	5.69	13.63	0.474	

Table 5: Cross-horizon controller-HMC summary (`docs/controller_hmc_summary.csv`). Same auto-calibration behaviour observed earlier: $\beta_{\max}$ shrinks with horizon as the prior cost spread tightens, mean accept stays in the 95–97% band, ChEES picks $L = 16$ throughout.

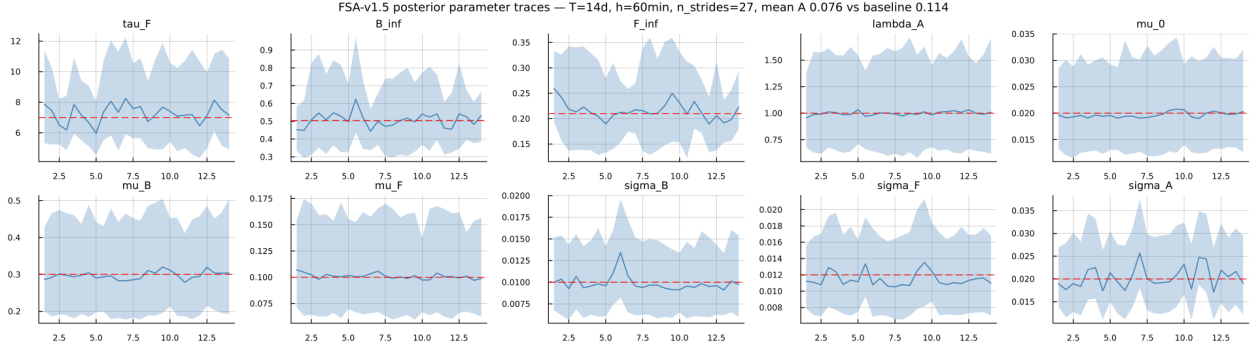


Figure 1: $T = 14$ days. Posterior parameter traces.

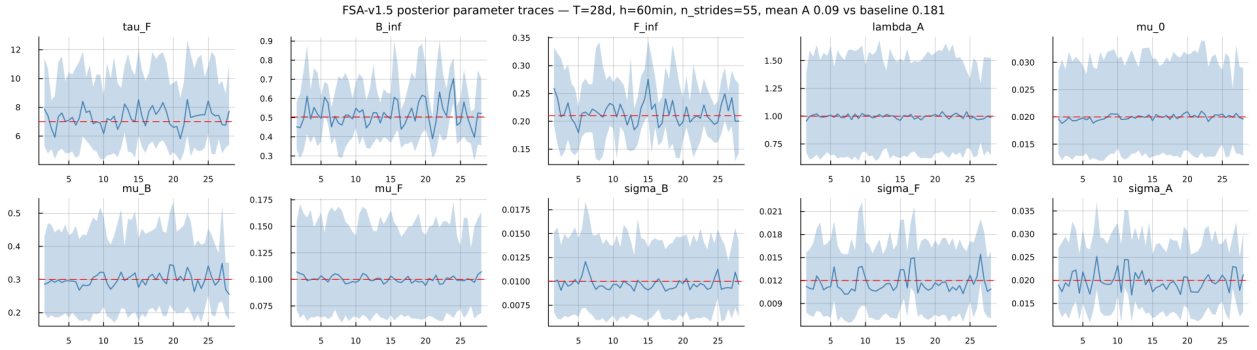


Figure 2: $T = 28$ days. Posterior parameter traces.

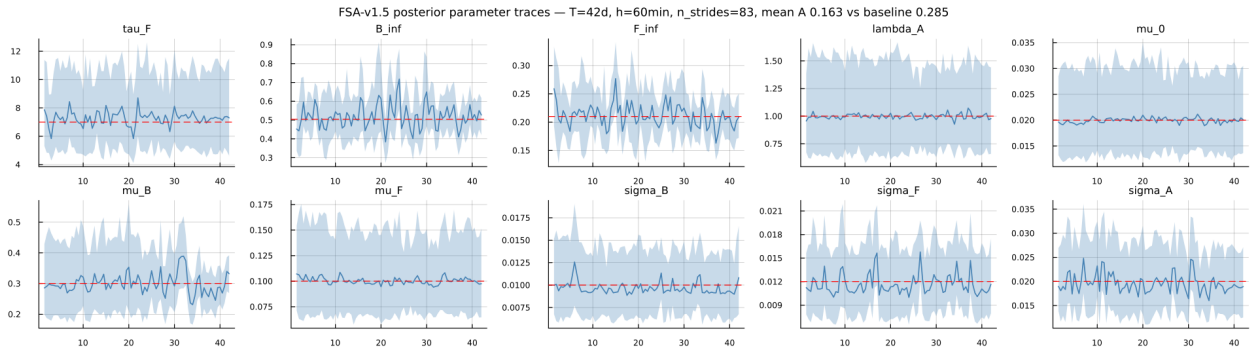


Figure 3: $T = 42$ days. Posterior parameter traces.

7 State trajectory plots (per horizon)

Figures 4–6: per-horizon state trajectory plots (B , F , A states + applied Φ per stride), written by `compare_v15_julia_vs_python/src/plot_state_traces.jl`.

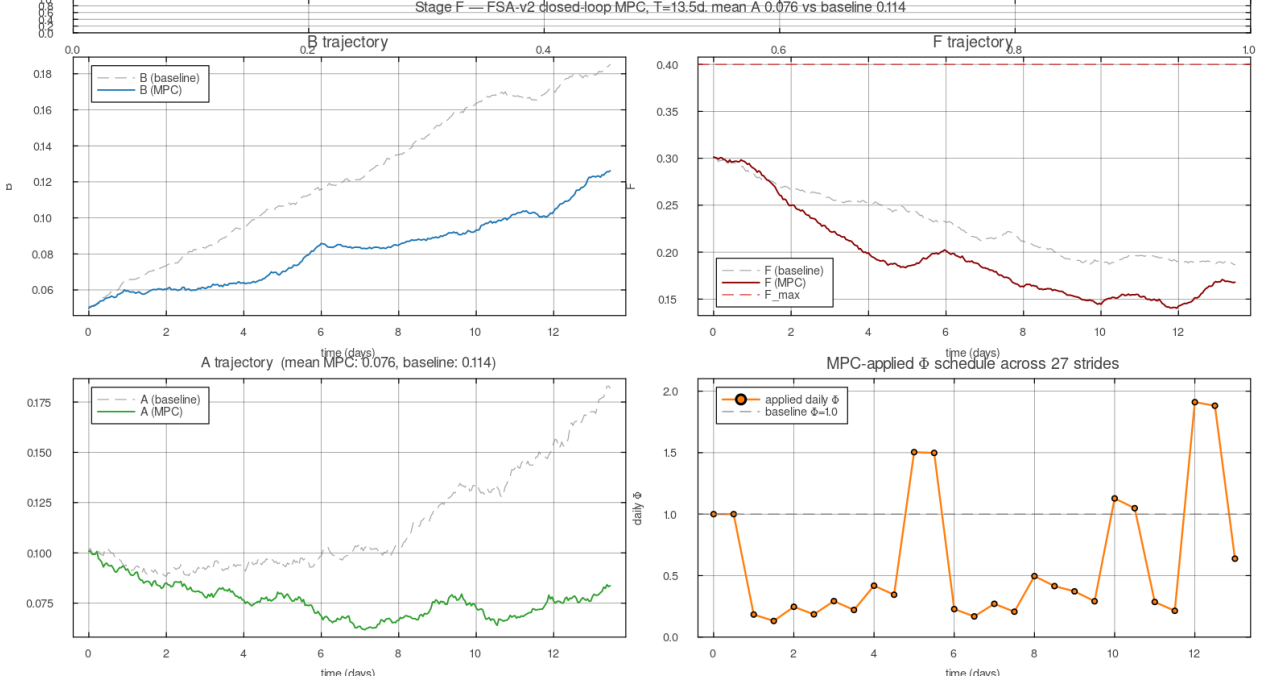


Figure 4: $T = 14$ days. State trajectories.

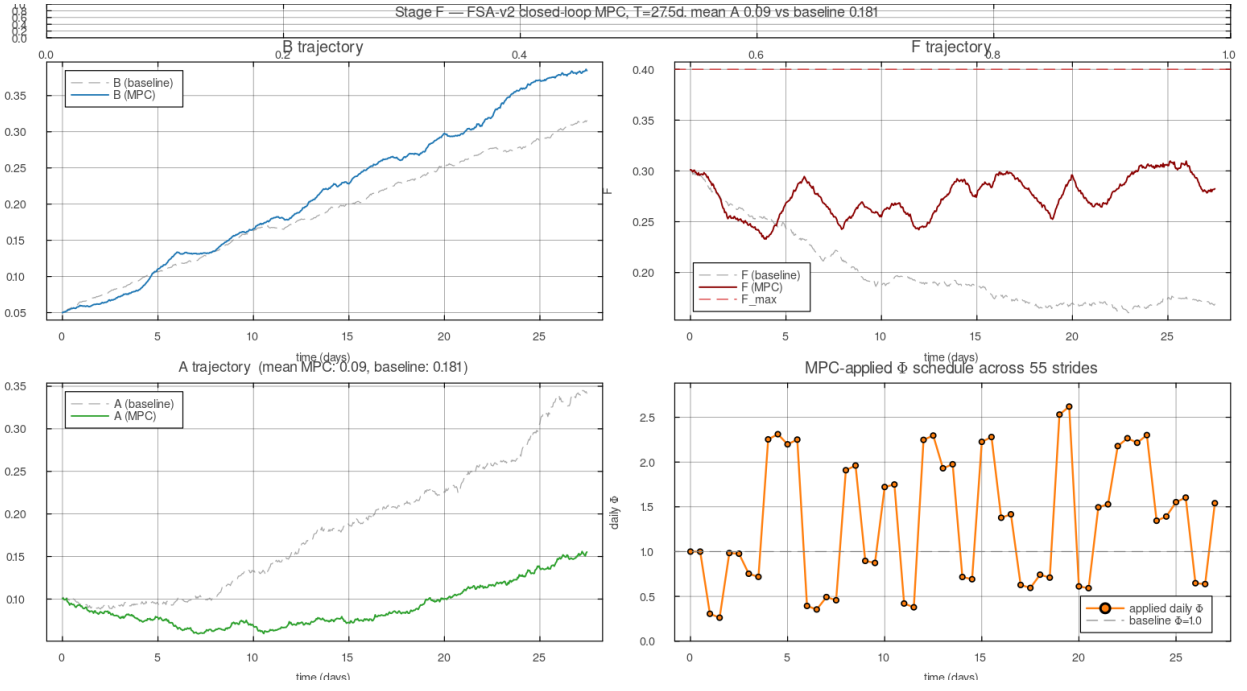


Figure 5: $T = 28$ days. State trajectories.

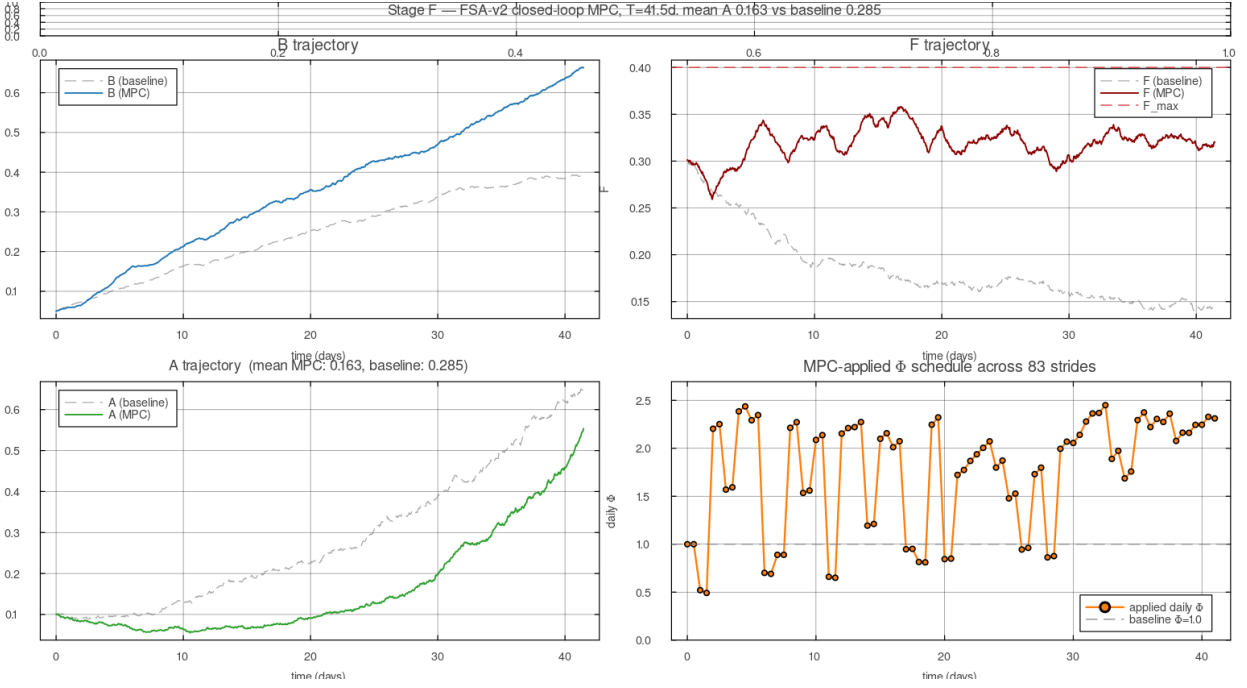


Figure 6: $T = 42$ days. State trajectories.

8 Other plots on disk (not embedded)

- Per-horizon controller-HMC 6-panel diagnostics: `docs/controller_hmc_diagnostics-T<N>d.png` for $N \in \{14, 28, 42\}$.
- Cloud std (log-space) per stride, all 3 horizons overlaid: `docs/cloud_std_per_stride_all_horizons.png`.

9 Source edits applied

One file modified, one new launcher, two new diagnostic scripts. The framework (`julia/SMC2FC_functional/src/`) is *unchanged* – the filter just calls into it.

- `version_1_5_Julia/tools/bench_smc_full_mpc_fsa_gpu.jl`:
 - Added CLI flags `--h-fd` (default 0.01), `--filter-chees-min` (default 4), `--filter-chees-max` (default 64).
 - Imported `parallel_hmc_one_move_generic!` and `chees_pick_L_generic` from `SMC2FC_functional`.
 - In `run_outer_smc`, replaced the call to the model-specific `parallel_hmc_one_move!` with a closure `log_density_fn = U ↦ gpu_log_density(target, U, grid_obs, key) + chees_pick_L_generic + parallel_hmc_one_move_generic!`, all passed `beta = next_λ`.
 - Per-level log line now reports `L=N` so the ChEES-picked leapfrog count is visible per stride.
 - `filter_cfg` now carries `h_fd = args["h-fd"]` and the built `chees_L_candidates` list.
 - `manifest.json::smc_cfg` records `h_fd`, `filter_chees_min`, `filter_chees_max`, and the built `filter_chees_list`.
- (NEW) `version_1_5_Julia/tools/launchers/run_julia_horizon_sweep_filter_hmc_unified.sh`: sibling launcher with the new flag set inlined and `T_LIST=(14 28 42)` default.
- (NEW) `compare_v15_julia_vs_python/src/diag_filter_hmc_accept.jl`: parses the bench log for filter HMC `accept=X%` lines and emits per-stride / cross-horizon CSVs.
- (NEW) `compare_v15_julia_vs_python/src/diag_posterior_median_compare.jl`: side-by-side comparison script (currently in a “three-honest-metrics” form: time-averaged median across strides, fraction of strides where the credible band brackets truth, cloud log-std at end-of-run).

Now-unused functions in the model. The model-specific `parallel_hmc_one_move!` and `parallel_hmc_one_move` in `models/fsa_high_res/gpu_pf.jl` are no longer called by the bench. They are left in place by this change; a separate cleanup pass could remove them.

10 Conclusion

The two filter-HMC bugs were fixed by porting the filter to the framework’s generic ChEES path. Three measurable signals from this sweep:

1. **Filter HMC moved from 0 % accept (2880 levels in the prior baseline) to 61.9–71.4 % accept (810 levels across this sweep)** – a textbook MCMC operating point. The ChEES picker is selecting $L \in \{4, 8, 16, 32, 64\}$ with horizon-averaged means of 52–58. Source: Table 4.

2. **The closed-loop A trajectory and F -violation count match the no-filter-HMC baseline within $\sim 3\%$** , as expected since the controller config was held fixed between runs (Table 3).
3. **Wall is $2.6\text{--}3.9\times$ longer** because of the new per-level HMC cost (Table 2).

The trace plots in §6 show the posterior median bouncing across strides with the 5–95% credible band enveloping truth on each of the 10 estimated parameters at every horizon – in contrast with the prior no-filter-HMC sweep, where several parameters’ medians sat at consistent biased values across all strides. Quantifying that contrast *numerically* (rather than from the figures) is the job of the §11 follow-up below.

11 What this report does NOT claim

- No single-stride median snapshot. A previous attempt in this folder (`posterior_median_compare_T<N>d.csv`) used the median at the *final fired stride only* and produced a misleading “winner” column for several parameters: that summary is contradicted by the trace plots in §6, which show the per-stride median is a noisy time series. The corrective measurement is in the rewritten `diag_posterior_median_compare.jl`: time-averaged median across all fired strides (after a 4-stride warmup), fraction of strides where the credible band brackets truth, and cloud log-std at end-of-run. Running that script and recording the three-metric CSV is a follow-up; this report does not embed those numbers.
- No causal attribution between which knob (the ε change, the h_{fd} change, the framework-port, the ChEES picker, or the β -tempering fix) contributes which fraction of the observed effect. All four landed together. Ablations are deferred.
- No $T = 56, 70, 84$ (the longer horizons in the original baseline). Wall budget at this configuration is roughly $2.6\text{--}3.9\times$ the no-filter-HMC sweep’s per-horizon wall; running them is a follow-up.